



Tribhuvan University
Faculty of Humanities and Social Sciences
Ratna Rajyalaxmi Campus
Pre-Board Examination 2025

R.N.

Bachelor in Computer Applications
Course Title: Web Technology
Code No: CACS 205
Semester: III

Full Marks: 60-40
Pass Marks: 24-16
Time: 3 2 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6×5= 30]

2. What is n- tier Architecture? Explain about three-tier architecture in detail. [2+3]
3. What are CSS properties? Explain CSS Box model along with its related properties and example.[1 + 4]
4. What is XML validation? Differentiate between XSD and DTD validators. [2+3]
5. Discuss about important tags used to create a table in HTML. Create a table using HTML that has a column: college, phone no. and 'logo'. In the last column, logos with hyper -links to respective websites should be placed. [2+3]
6. Explain about any five types of CSS selectors with their syntaxes. [5]
7. What is server-side scripting? Write a server-side program to take two numbers from HTML form and display the sum of them in browser. [1+4]
8. Explain about error handling techniques in server-side program with example. [5]

Group C

Attempt any TWO questions

[2×10 = 20]

9. Compare HTML, CSS and XML technologies. How XML documents are linked with DTD files? Write an XML code to store following information and also create XML Schema specifying these criteria. [4 + 6]
 - Each student has a name, address, phone and website element.
 - Address might appear multiple times
 - Address has attribute named "type" with value permanent and temporary
 - Phone must be 10 digits.
10. Create a web application to design a form using HTML and CSS to take details of a product for listing in an e-commerce site. Now write a server-side program to store the detail of the product in a relational database table, *products (id, name, description, price, quantity)*. When user clicks on 'save' button in the form, the entered data should be inserted into the table. [4+6]
11. What is importance of database in web application? Write a program to display all records of student in a web page in tabular format after reading from a database table using server-side scripting. (Assume the table contains some records of students). Also include a button for each record in HTML page and include the functionality to delete the corresponding record from database when the button is clicked. (Assume your own database, database connectivity constrains and table structure.) [2+8]



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Pre-Board Examination 2025

Bachelor in Computer Applications

Course Title: OOP in Java

Code No: CACS 204

Semester: III

Candidates are required to answer the questions in their own words as far as possible

Full Marks: 60-40

Pass Marks: 24-16

Time: 3 2 hours

Group B

[6×5= 30]

Attempt any SIX questions.

11. Create a method inside a class that takes an array of string as parameter and returns true if lengths of length of every string inside it is even and returns false otherwise. Create object of that class and call the method in main method and print the result. [5]
12. What is thread life cycle? Write a program to demonstrate multithreading by creating thread using Runnable interface. [2+3]
13. Differentiate between *throw* and *throws* keyword in Java. Write a program to demonstrate custom event handling in Java. [2+3]
14. How can you create menu and submenu in swing? Write a suitable program to create a frame with menus and submenus. [2+3]
15. Write a Java program to create a base class *Vehicle* with attributes brand, speed, and type, and a method called *describe()* that prints out its brand and type. Inherit the class in Car and Bike classes and override the *describe()* method to print specific messages for each vehicle. [5]
16. How can we use listener interface to handle events? Write a program to demonstrate the handling of button click event. [2+3]
17. How byte stream and character stream are different? Write a Java program to read the contents of one image file and write to another image file. [2+3]

Group C

[2×10 = 20]

Attempt any TWO questions.

18. How static polymorphism and runtime polymorphism is different? Write a program considering following scenario:

You are to develop a Payment Processing System for an e-commerce platform that can handle various payment methods, including Credit Card, PayPal, and Bank Transfer. Although each method processes transactions uniquely, they all share a common *processPayment* operation. To achieve this, you must define an interface named *IPaymentMethod* containing a single method, *processPayment(double amount)*. Subsequently, create three distinct classes—*CreditCardPayment*, *PayPalPayment*, and *BankTransferPayment*—each of which will implement the *IPaymentMethod* interface and provide its specific implementation of the *processPayment* method; for example, *CreditCardPayment* should indicate it's "Processing credit card payment of Rs. [amount]...", *PayPalPayment* should state it's "Initiating PayPal transaction for Rs. [amount]...", and *BankTransferPayment* should confirm it's "Processing bank transfer for Rs. [amount]...". Finally, within your *main()* method, you will demonstrate runtime polymorphism: based on user input (e.g., selecting 1 for Credit Card, 2 for PayPal, or 3 for Bank Transfer), an *IPaymentMethod* reference will dynamically point to the chosen payment object, and its *processPayment* method will be invoked for a given transaction amount, simulating how an e-commerce system adapts to different payment methods. [3+7]

19. How adapter class and listener interface are different? Explain the steps to connect and access database table using JDBC. Write a program using JDBC to insert a record into a database table and display all the record of that table in console. (Make your own assumption about database server, database and table structure). [2+3+5]
20. Explain about any two layout managers in GUI development using swing with suitable programming example for each. Write a program to take name and phone number values from a swing frame and store the values in a database table when save button of that GUI is clicked. (Make your own assumption about database server and database). [4+6]



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Bachelor in Computer Applications
Course Title: Probability and Statistics
Code No: CAST202
Semester: III

Full Marks: 60-40
Pass Marks: 24-16
Time: 3 2 hours

Candidates are required to answer the questions in their own words as far as possible.

Group – A

Attempt any FOUR questions: (4×5 = 20)

2. Define statistics. Write the role and limitation of statistics in computer application.
3. Calculate the mean, median and mode of the following data:

Age(in year)	20-29	30-39	40-49	50-59	60-69
No. of persons	6	9	12	7	6

4. Define correlation. Calculate the pearson's correlation coefficient and interpret the result:

Age	20	30	40	50	60
Blood pressure	115	120	125	135	140

5. Define regression. Following data shows the age in month and weight in kg of newly born baby:

Age	0-	3	5	6	8
weight	2.5	4	5.5	8	10

Estimate the weight of the baby at the age of 4 months.

6. What do you mean by the binomial distribution? The mean and S.D. of a binomial distribution are 12 and 3 respectively. Find the probability of exactly 2 successes.
7. Write the formulae to estimate the sample size for population mean and population proportion. What should be the sample size so that the sampling error may be ± 3 in 95% confidence interval and population standard deviation is 8?
8. List the probability and non probability sampling. Techniques. Write in brief about the systematic random sampling.

Group C

Attempt any ONE questions:

(1×10 = 10)

9. Two brands of computer are tested for their life and following results were obtained:

Life(in year)	2-4	4-6	6-8	8-10	10-12
Brand A	10	12	18	14	8
Brand B	8	12	16	10	5

Both brands are offering the same price and advertising in favor of their brands saying that the brand has consistent life. If you are required to purchase computer of one of these two brands, which computer would you prefer and why?

10. Given a normal distribution with mean 65 and variance 100, find the probability that

(a) $P(X < 75)$ (b) $P(X > 60)$ (c) $P(50 < X < 70)$ (d) $P(X < 60 \text{ or } X > 70)$
 (e) 40% of the values are less than what value of X?

11. A computer engineer wants to know the average downloading time in seconds to download the image file from four different computers. Using the following information provided below, use appropriate statistical test whether the average downloading time is significantly different or not at 5%:

Computer I	Computer II	Computer III	Computer IV
22	26	30	18
18	24	25	15
12	12	20	11
14	10	15	16

Best of Luck

Group B

Attempt any six [5 x 6 = 30]

2. What is Data Structure? Explain asymptotic notations with example.
3. What is stack? Write an algorithm to perform PUSH and POP operation in stack
4. Describe Prim's algorithm to solve MST problem with suitable illustration.
5. Write are the limitation of linear queue? How circular queue can used to overcome the limitation of linear queue. Describe enqueue and dequeue operation in circular queue.
6. What is divide and conquer algorithm? Explain collision resolution techniques with suitable example.
7. Describe radix sort algorithm with example.
8. Create Huffman Tree and calculate Huffman code for the following characters along with their frequencies using Huffman algorithm.

Characters	A	E	I	O	U	S	T
Frequencies	10	15	12	3	4	13	1

9. A binary tree T has 9 nodes. The inorder and preorder traversals of T yield the following sequence of nodes:
In-order: E A C K F H D B G
Pre-order: F A E K C D H G B
Draw the tree T.

Group C

Attempt any two [2 x 10 = 20]

10. What is linked list ? Describe types of linked list. Write an algorithm to insert and delete node in following position of doubly linked list.
 - at the beginning of doubly linked list.
 - at the end of doubly linked list.
 - at the specified position of doubly linked list.
11. Define min heap and max heap. Explain heap sort algorithm and trace it to sort the data:
25 67 56 32 12 96 82 44
12. Differentiate between BST and AVL tree. Construct AVL Tree from following sequence.
14, 17, 11, 7, 53, 4, 13, 12, 8, 60, 19, 16, 20



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Bachelor in Computer Applications
Course Title: System Analysis and Design
Code No: CACS203
Semester: III

Full Marks: 60-40
Pass Marks: 24-16
Time: 3 2 hours

Candidates are required to answer the questions in their own words as far as possible.

Group – B

Short Answer Questions 4*5=20

Attempt any 6 question.

1. Explain different types of testing with example.
2. Explain different requirement collection technique.
- 3 List the advantages of Normalization. Explain 2NF with an example.
4. What are the key features of a project manager?
5. What do you understand by feasibility? Explain all the types of feasibility.
6. Briefly explain the different types of maintenance.
7. Explain the factors to be considered while designing a interface?

ANALYSIS
DESIGN
ABDIM

Group C

Long Answer Questions 1*10=10

Attempt any 2 question

8. What are the core tasks involved in the systems development process? Identify and describe some common issues associated with the traditional Waterfall Software Development Life Cycle (SDLC). Additionally, provide a brief explanation of Agile development methodology.
9. What does "balancing Data Flow Diagrams (DFDs)" refer to? Create a context diagram and detailed data flow diagrams for a student management system.
10. A bank has the following policy on deposit. On same deposit for a period of Rs. 50,000 and above for five years or above the interest rate is 15%. On same deposit for a period less than 5 year it is 12%. On deposit below Rs.50,000 the interest is 10% regardless of period of deposit. Write the above process using:
 - a) Structured English
 - b) Decision Table

Sci